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SIMATS ENGINEERING
ASSESSMENT TOOL 2
Case study

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions

Assessment Tool Weightage: 40%

Code of Conduct:

I AAMU (192524257) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

AAMU
Signature of the student:

QUESTIONS

1. A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an 8% packet loss, 180 ms jitter, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.

Questions

1. Analyze whether the problem is primarily caused by the transport protocol (TCP/UDP) or the application layer.
2. Justify your answer based on the communication requirements of real-time multimedia applications.
3. Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.

2. A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the average DNS lookup time is approximately 800 ms, significantly affecting user experience. The organization plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.

Questions

1. Analyze the reasons behind the high DNS resolution time.
2. Propose a suitable DNS architecture using techniques such as Anycast DNS, DNS pre-fetching, and TTL optimization to reduce the lookup time below 50 ms.
3. Evaluate the advantages and limitations of your proposed architecture in terms of performance, scalability, and availability.

3. A cloud service provider delivers applications to millions of users worldwide. During promotional events, users experience intermittent delays in accessing cloud-hosted services. Investigation reveals that DNS queries are taking nearly 800 ms because all requests are directed to a single primary DNS server. The company wants a highly available DNS solution capable of handling global traffic efficiently.

4. Examine the impact of centralized DNS architecture on application performance.
5. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.
6. Analyze how TTL values, Anycast routing, and DNS caching influence system performance and fault tolerance.

4. A financial institution operates an electronic stock trading platform where thousands of buy and sell orders are submitted every second. During market opening hours, the average order acknowledgment latency increases from 1 ms to 40 ms, resulting in delayed trade confirmations and customer dissatisfaction. Network engineers observe increased TCP retransmissions, longer connection establishment times, and excessive buffering.

Questions

1. Analyze three possible TCP-related factors responsible for the latency increase, considering **flow control**, **Nagle's algorithm**, and **SYN queue limitations**.
2. Explain how each factor affects transaction processing in high-frequency systems
3. Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity, difficult to understand

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Video Conferencing Performance Analysis

Case understanding:

The multinational (IT) company is experiencing voice distortion, frozen video frames, and delayed communication during peak business hours. The network monitoring data shows 8% packet loss, 180ms jitter, & fluctuating bandwidth utilization. The administration suspects either the transport protocol or the application-level buffering strategy.

- i) Analyze whether the problem is primarily caused by TCP/UDP or the application layer:

The problem is primarily related to network condition and the transport/application strategy, rather than being caused by UDP alone.

For real-time video conferencing, UDP is generally more suitable. TCP provides reliable, ordered delivery through acknowledgment & retransmission. However, when packets are lost, retransmission. However, when packets are lost, retransmission can cause additional delay.

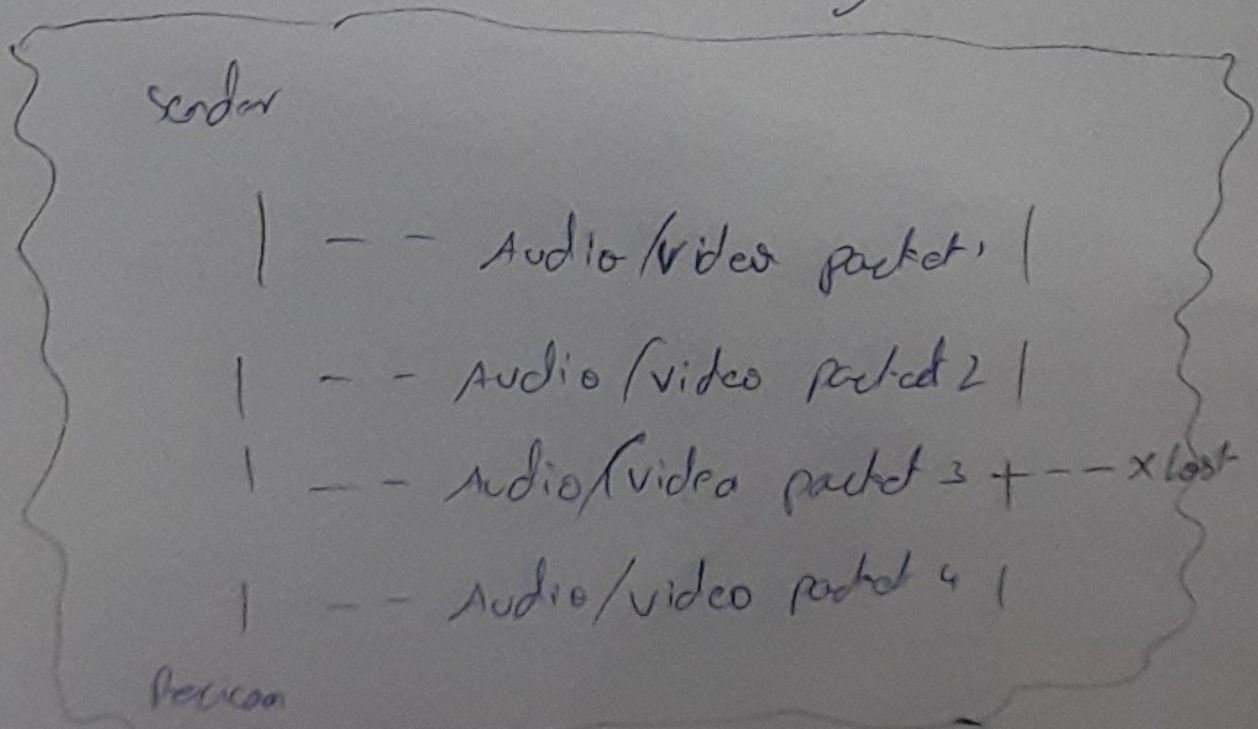
The loss of packets and the packet loss is particularly serious for real-time communication. Simply changing TCP to UDP would not completely solve the problem. The application must also handle packet loss, jitter, & changing bandwidth appropriately.

A) Justification Based on Real-Time multimedia.

UDP Advantages:

*) UDP does not require connections established acknowledgement or retransmission of every lost packet.

*) Therefore, it can be delivery packets with lower overhead and lower latency.



a) we UDP-Based Realtime Transport

we use UDP for real-time Audio/Video traffic where appropriate, with application-level mechanisms to manage packet loss and jitter.

B) Adaptive jitter buffer:-

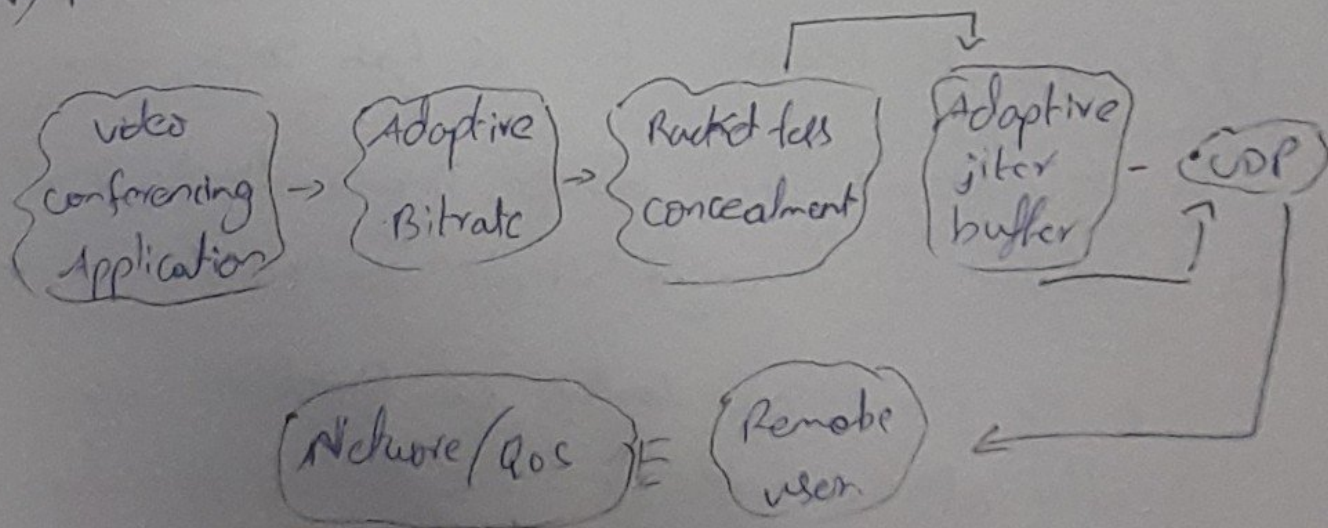
The conferencing applications should dynamically adjust its jitter buffer.

- If jitter increase $\rightarrow$ Temporarily increase buffer size

- If network condition improve $\rightarrow$ reduce buffer size.

- Avoid Excessively large buffer because increase delay

iv) Recommended architecture:



iii) Targeting DNS lookup delay cons.

The exact improvement will depend on geographic placement, network, below 50 ms is a good target, not a guaranteed result from the same information alone.

iv) Advantages of proposed Architecture:

- performance: Any cost-reduce geographic network.
- scalability: multiple DNS can handle requests simultaneously instead of depending on one location.
- Availability: If one DNS location becomes unavailable, traffic can be routed toward another DNS location.

v) Limitations:

- Poorer careful routing & monitoring.
- cached records can become outdated.
- May generate unnecessary queries.
- can delay propagation of DNS changes.

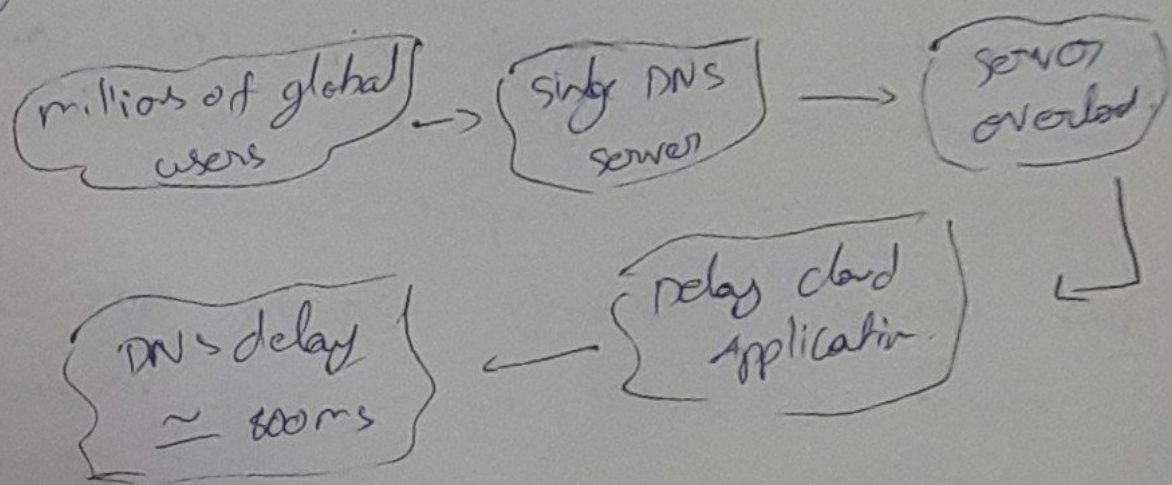
vi) Conclusion:

The slow DNS lookup is primarily caused by inefficient DNS resolution infrastructure, such as geographic distance, inadequate caching, suboptimal and lack of optimization.

Centralized DNS Solution for Global cloud services

The case states that a cloud service provider serves millions of users worldwide, but during promotional event DNS queries take all requests are sent to a single primary DNS server. The organization therefore needs a highly available.

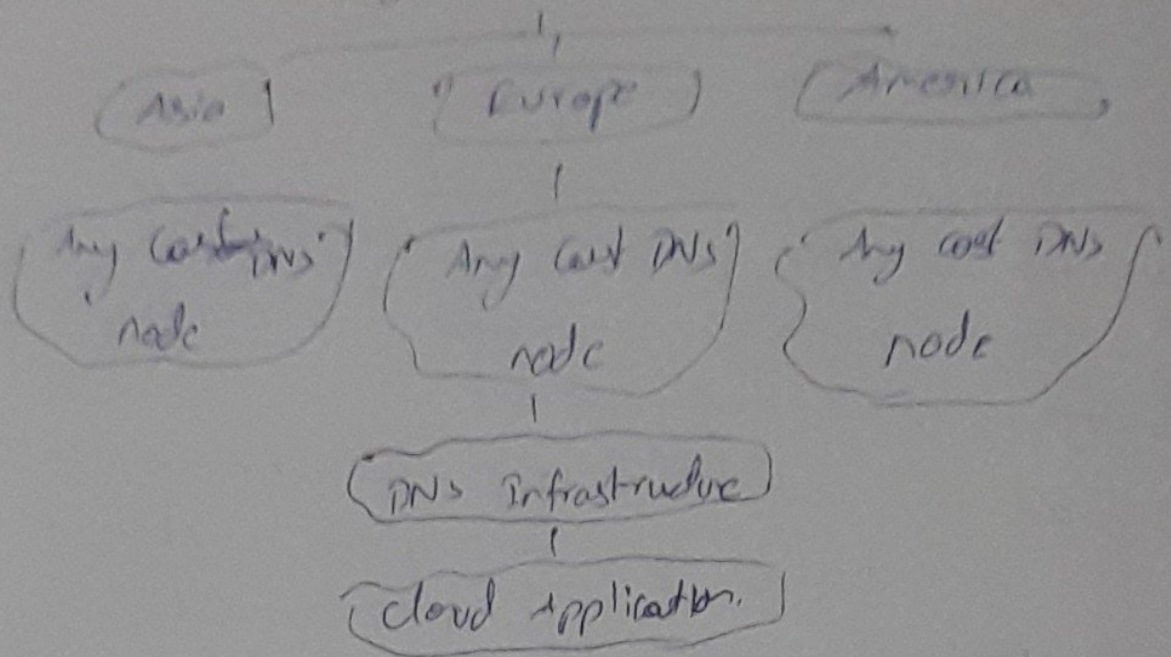
i) Impact of centralized DNS architecture application
+ centralized DNS architecture creates a single point of dependency for DNS resolution.



ii) Design of a Distributed DNS Solution:

The recommended solution is a distributed Anycast DNS architecture with DNS caching appropriate values, and multiple DNS locations.

global view



A) Anycast routing: with Anycast multiple DNS server can advertise the same DNS services & addresses from different geographic location.

B) DNS Caching: frequently requested DNS records should be cached by recursive resolvers.

C) Multiple DNS servers: Instead of depending on one primary server, the organization should maintain multiple nodes across ge.

iii) Influence of TTL, Anycast Routing & DNS:

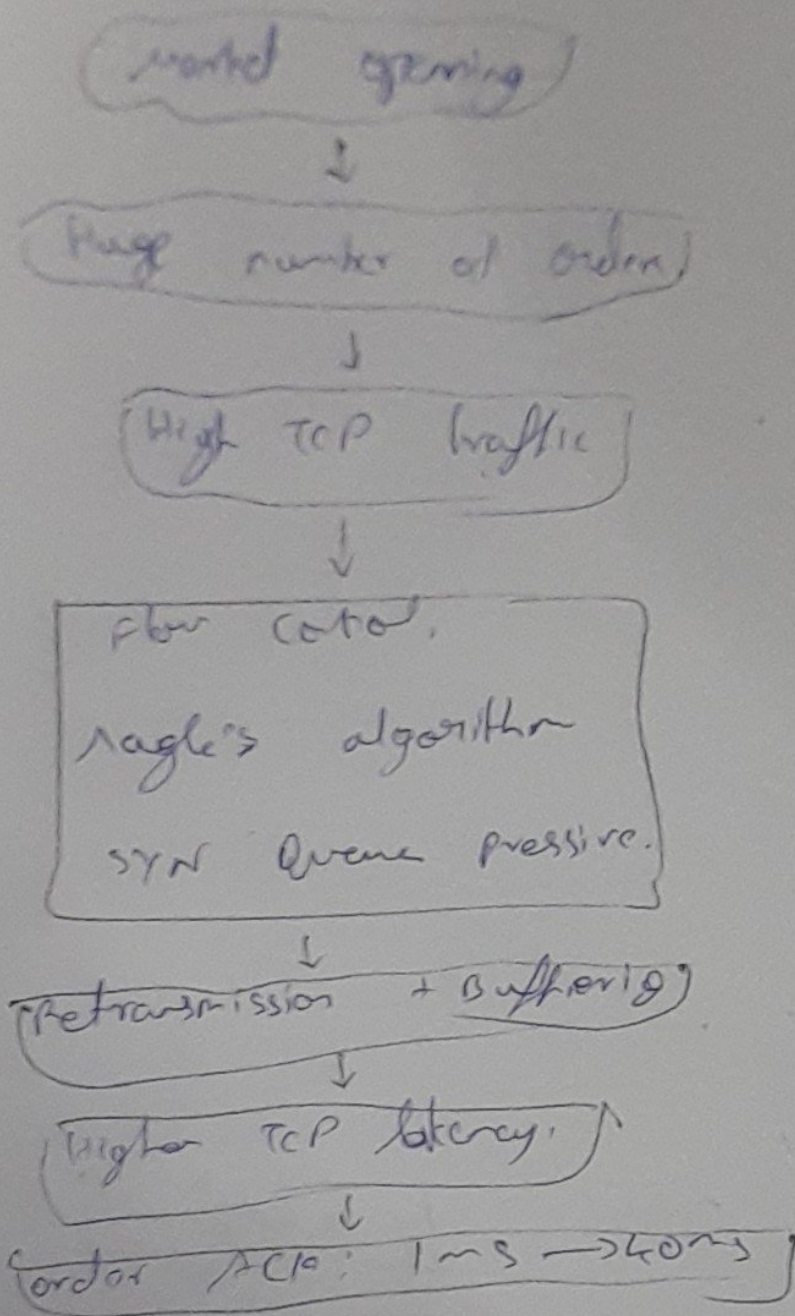
1) TTL values: TTL (Time to Live) controls how long a DNS record remains in a cache.

Q) TCP latency analysis in an electronic stock trading platform handle thousands of buys and sell orders. During opening hour, latency increases from 1 ms to 100 ms. Network engineers also observe increased TCP layer connection establishment times and excessive buffering.

i) Three TCP-related factors responsible for latency increase.

A) TCP flow control: TCP uses a receiver's window to prevent a sender from transmitting more data receiver can currently handle. During heavy activity receiver or application may data more slowly. The advertised receiver window can decrease, causing the sender to slow down.

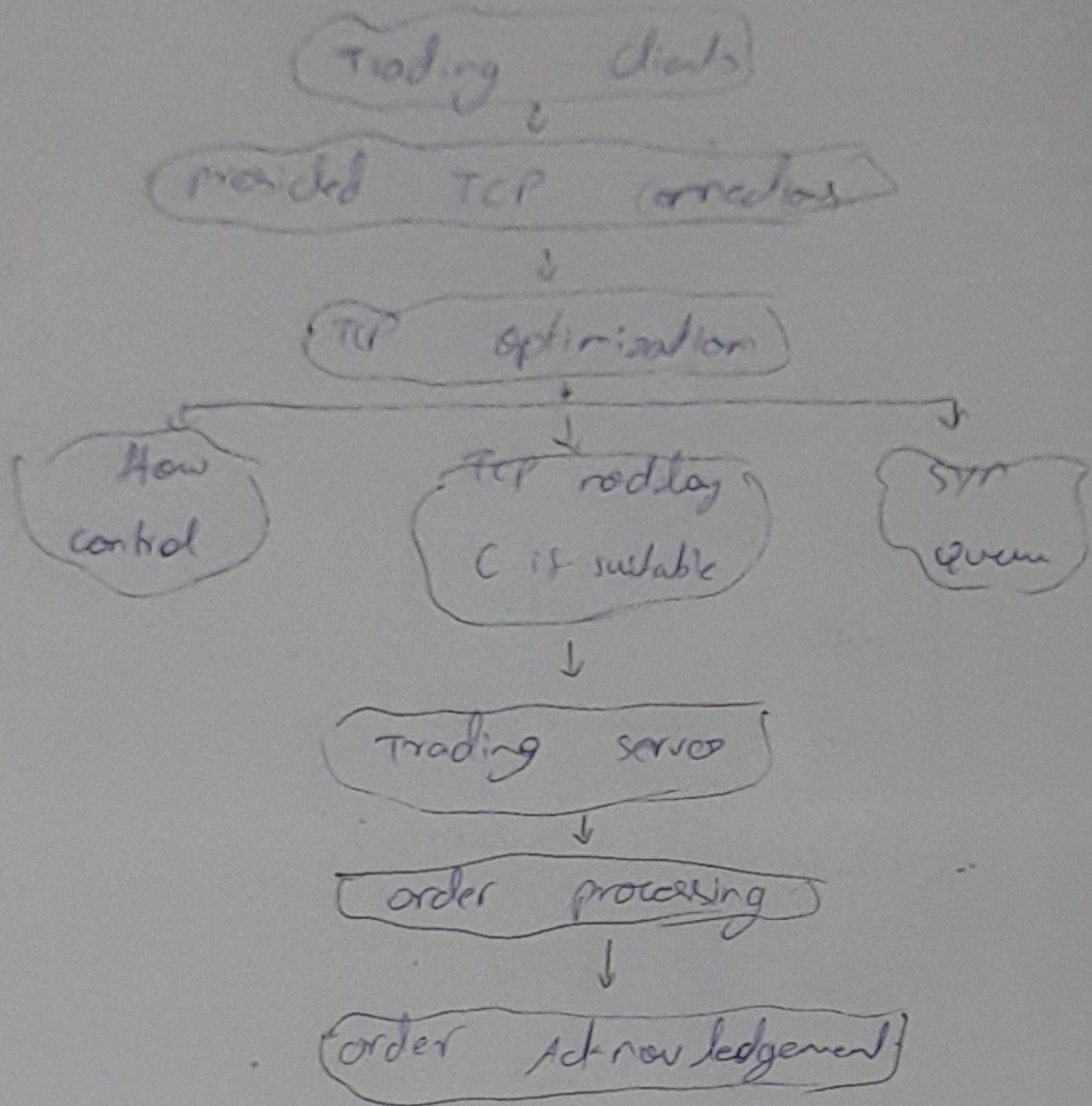
b) Nagle's Algorithm: Nagle's algorithm attempts to reduce number of small TCP segments by combining small amount of data sent together. The cost is useful reducing overhead but can be introduced delays into application sends small message.



(i) Recommended TCP Configuration:

- * optimized flow control,
- * reduced nagle-related delay.
- * Increase/optimization SYN Queue capacity
- * Reduced retransmission.

ii) Recommended Architecture



✓) Conclusion:

They increases in order of acknowledgement from 1ms to 40ms is consisted with combination of TCP controls.